Toward a four-dimensional Yang–Mills construction

Notebook 01: proof obligations and verified reductions

Research working document

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Abstract

The goal is a complete existence argument for a nontrivial four-dimensional Yang–Mills quantum theory. A first intermediate target is a nonzero, fully Euclidean-covariant continuum observable sector obtained from one subsequence of four-dimensional $SU(2)$ Wilson theories. Cutoff uniqueness is unnecessary for that target. This notebook gives explicit conditional reductions for compactness, factorial correlation bounds on joint flat tests, rotation restoration, and positive-energy nontriviality. Fixed-side patch grids repair the all-order geometry conditionally on uniform translated-grid source estimates. An independent free-lattice calculation reduces one nontriviality test to fixed relative kernel accuracy. It also gives counterexamples to several tempting shortcuts. The model-specific source estimates, the interacting rotation input, and the passage from one scalar sector to Yang–Mills remain open. Companion Notebook 02 now gives a fixed-coupling proof preserving diagonal reflection positivity in the original coordinate-torus infinite-volume limit, with the original odd-L regulator. It supplies no uniform ultraviolet estimate. Companion Notebook 03 proves the local current commutator with explicit constants and a convergent Gaussian sharp-cutoff cluster expansion. Its remaining interfaces are the combined current at complex decoupling parameters and strict locality in the background variables. For the Gaussian cutoff part of the quadratic coefficient, it also proves a direct Hessian bound and volume-limit criterion that avoid strict background-support activities, conditional on actual uniform source-response estimates and their identification. Notebook 04 establishes strict background locality of the Gaussian cutoff logarithm under a uniformly small local precision perturbation. Matching that hypothesis, its domain and its symmetries to the actual Yang–Mills Gaussian is still required. Notebook 05 proves a quantitative flat axial-to-exponential gauge bridge with the actual averaged-path penalty, and an interacting cutoff theorem for disjoint fixed-dimensional cells. It also identifies the correct physical-field radius estimate and the full exponent's linear Jacobian term. Notebook 18 extends that integration theorem to overlapping supports under an explicit weighted local coefficient norm. Notebook 06 supplies direct scale-uniform flat quadratic bounds through curvature averaging, a finite-block axial gap, and an explicit compact bounded-energy right inverse. Their homogeneous identification with the constrained-minimizer Hessian now follows from the nonlinear construction of Notebook 15 and the explicit match in Notebook 17. Notebook 07 proves exponential locality of this flat action and retained covariance, and their refinement limit at rate $O(M^{-2})$ in an exponentially weighted kernel norm, uniformly in compatible periodic volume. Its explicit sinc-alias limit is a free Gaussian reference. Notebook 19 proves the physical homogeneous complex-background perturbation in the exact source coordinates; the interacting continuum limit remains separate. Notebook 08 constructs the corresponding fine minimizing extension in the deposited homogeneous flat source gauge. It proves uniform exponential locality for the potential, first fine derivatives, curvature, and componentwise Laplacian, with physical Hölder bounds below the endpoint for the first derivatives. These local flat response maps are inputs to a further perturbation argument; they do not establish that argument on a nonflat current domain. Notebook 09 proves locality of the full flat fine inverse, with two total fine derivatives in physical cell Hilbert norm, and a refinement-uniform complex neighborhood for the deposited auxiliary current. Its coarse precision has a quadratic zero-current remainder and all-order localized derivative bounds. Notebook 10 identifies a local projection onto feasible curvature variations and gives a full weighted expansion for a small curvature metric perturbation

at fixed averaging constraint. Notebook 11 bounds the separate nonlinear constraint-current term by composing the corrected fixed-L quadratic jets and summing their physical scale factors. It establishes a common weighted Hilbert domain for the full flat current family along $M = L^j$, including factorial current derivatives. Nonflat and retained-history domains remain separate. Notebook 12 strengthens the flat inverse to arbitrary-forcing pointwise $C^{1,\alpha}$ bounds for $\alpha < 1$, and transfers the full flat current to the homogeneous source norms N, N_β on a smaller uniform complex ball. Its current requires only a supremum bound. It also proves the one-form right-divergence members and the separate constrained current member. For every fixed exponent in $(0, 1)$, an improved heat-integral argument removes the Hölder loss. A scalar proof requires an explicit uniform positive penalty interval; the baseline Laplacian estimate is not asserted for nonzero current. Notebook 13 proves a uniform analytic ball for the complete fixed-reference corrected nonlinear average and an exact local chart of its constraint. The chart has a uniformly local right inverse for its ordinary differential. Notebook 14 computes the exact change to the moving-reference derivative, including its endpoint correction. A cumulative gauge relation controls the ordinary reference trajectory and the actual bond and scalar averages. A Wilson plaquette calculation then gives a uniformly local energy inverse on a small global physical C^1 ball, including the moving gauge projection. Its bare Wilson action, physical pairing and left-exponential Hessian chart match the deposited conventions in dimension four. Original-paper verification, the precise graded-curl and effective-Hessian identifications, nonflat pointwise/current estimates and retained-history domains still require their separate checks. Notebook 15 closes a nonlinear classical minimum construction in the homogeneous small-global-field class. An equal-exponent constrained inverse, the exact Wilson force, and a two-Hilbert constraint-curvature estimate give a uniform analytic fixed-slice minimum. A genuine constraint-preserving gauge flow and a separate ordinary-average transfer produce a minimum modulo gauge for the ordinary homogeneous constraint. Its radii and physical $C^{1,\alpha}$ bounds are uniform in refinement and volume. This does not supply mixed histories, large-field integration, or the interacting continuum observable hierarchy. Notebook 16 proves exponential locality of both selected minimizer representatives and their multiplier, at small nonconstant backgrounds, with one weighted direction at every analytic order. The exact action-envelope identity gives a local nonlinear classical coarse Hessian, its small weighted perturbation from q_M, and a classical coarse gauge identity. A positive auxiliary coarse penalty gives a stable local inverse. Quantum fluctuation integration and multi-insertion tree estimates are not supplied by those classical response bounds. An exact constrained-Gaussian trace calculation in Notebook 16 also shows that its physical curl variance grows as η^{-4}. The probability of a fixed smooth ball for a deterministic background plus gZ is at most $Cr\eta^2/g$. Notebook 17 supplies that rescaling and joint local-tail analysis. It proves real correlated bad-block bounds and a complex marginal estimate without a total-volume factor, and compares the actual first-step reduced Wilson Hessian with the flat energy, including constraint curvature. It also identifies the homogeneous flat current-step covariance uniformly over all earlier averaging depths. Notebook 18 proves the overlapping-interaction cutoff expansion and its quantitative comparison with the Gaussian logarithm. Under a small local precision decomposition it also supplies strictly local background activities. Notebook 19 constructs the physical stationary background and its exact pivot-coordinate Gaussian, including both action curvature terms, uniformly in preceding classical depth. It also bounds the local Jacobian, Haar and gauge interactions. Notebook 20 closes the actual bare Wilson small-field integration: the action and precision have explicit local decompositions, the retained cutoff is fixed, and the logarithm has a volume-uniform connected expansion. Its background-only delta and Gaussian determinants are included separately. This support statement is in independent current backgrounds, before composition with the next-coarse stationary field. Notebook 21 proves strict support localization of the homogeneous classical minimizer and action uniformly in earlier refinement. A rooted support contraction, with arbitrary fixed cardinality reserve, gives local scalar action coefficients and individually symmetric precision pieces. The physical chart substitution then closes one small-field integration of that classical reference action at every earlier classical depth. Notebook 22 constructs the actual stationary current background as a strictly supported analytic function of next-coarse data and transports the generated activities through it. That bounded step has explicit analytic-radius and support-weight losses; it does not give repeated-step closure. Notebook 23 proves local gauge covariance of the complete small-field correction, including the cutoff, Haar and pivot factors, and Gaussian determinant. Its total linear

jet vanishes and its quadratic jet obeys the finite-regulator Ward identity. Notebook 24 bounds the inherited fluctuation increment by $O(gR)$ divided by the available analytic margin and uses the exact pivot support geometry to give a sufficient next-step interaction-norm inequality. The resulting logarithm comparison is proved under that inequality. Notebook 25 gives a smaller pre-root-gauge representative with an inverse-ratio link-angle bound and a strict support series, uniform through iterated classical blocking. It extends the one-step composition germ of a gauge-invariant total functional where that smaller representative is admissible. Notebook 26 supplies independent auxiliary integration cells that satisfy the support-weight test: a fixed positive connected-carrier exponent is recovered through inherited insertion, Gaussian integration, all normalization terms and stationary transport. The physical blocking and cutoff stay fixed. This is a bounded step on a smaller domain, with explicit noncontractive constants. Notebook 27 constructs the actual composite ordinary blocking map, its local pivot chart and retained Gaussian, and the precisely normalized compact-gauge integral. Notebook 28 composes around a local core of the pre-gauge representative, expanding only its small distant tail. Classical support substitution before integration then gives one small-field update on the same input and output analytic ball, with a fixed positive carrier exponent and a nonvanishing Gaussian cutoff integral. It also proves an inverse-ratio background contraction for coefficientwise Taylor remainders of field degree at least five, and a contractive quantum response to input differences with matching low jets. The update can regenerate low jets and the general activity bound can grow. Notebook 29 proves the actual local spatial symmetries and classifies the total quadratic, cubic and quartic marginal vertices through their nonlinear Ward identities. One normalized gauge-invariant classical action supplies their common subtraction. Complete first-gradient Hölder support families and an exact lattice difference factorization give a connected-norm spatial gain $C(J)L^{-\alpha}$ under the stated infinite-kernel and coherent-family hypotheses. A classical semigroup modulo gauge permits $L = R^q$ to be chosen after fixed R, J. Joint compactness realizes the coherent infinite interactions at fixed parameters. A complete quotient by equality of normalized totals makes classical representation changes and bounded full-reference extraction precise; the resulting classical contraction retains its joint-map hypotheses. Notebook 30 closes the universal gauge identity for that fixed classical system and constructs a norm-preserving finite periodization of invariant interaction classes. Strict Gaussian local continuity, with explicit carrier-size and expansion-order tails, defines the actual homogeneous local quantum map on the quotient. It identifies varying finite inputs and every fixed finite history whose intermediate finite families meet the uniform budgets. Exact transport of the actual classical reference makes intermediate extractions telescope. A finite block of projected quantum steps has an $O(b^2)$ stable amplitude envelope while its couplings remain uniformly weak. A fixed outer domain supplies the matching $O(b^2)$ zero-input force, and each inverse-coupling increment has a bounded absolute size. Notebook 31 closes the finite representative-budget interface. Finite Ward repair and periodic localization retain an explicit wrapping error at the same field radius and carrier exponent. Exact finite quotient descent and equivalent small representatives then give uniform local induction through every permitted finite history while its couplings remain weak. All raw constants are compensated in the vacuum factor. A fixed-threshold parity argument isolates the pivot and Gaussian determinant as the leading zero-input correction. On the jointly compatible infinite interaction space, a quadratic-only spatial estimate at fixed radius proves exact classical preservation of the Maxwell coefficient. The leading quantum coefficient and its drift sign still require evaluation; the infinite identity does not discard finite cutoff-extraction errors. Notebook 32 proves qualitative uniform disappearance of that finite classical leakage along cofinal nested periods and joint continuity of the finite quantum map in its class and positive real coupling. It derives the actual leading determinant's convergent Hessian trace and a finite composition identity from one Haar pushforward evaluated directly and sequentially. Together with classical Maxwell preservation this gives additive infinite leading coefficients. An exact gauge/coarea identity retains vector, based-ghost and coarse-constraint determinants; a Rayleigh test rules out a uniform physical mass bound on the based ghost alone. Identifying the coarse constraint covariance with a shifted actual effective-Hessian inverse gives a uniform bound on that coarse determinant's Maxwell contribution for a fixed sufficiently large shift. A positive block margin uniform over all varying-depth structural limits would transfer to actual finite regulators by qualitative compactness, without a rate. A separate shooting theorem tunes arbitrarily long finite local histories to a fixed weak endpoint under an explicitly unproved positive averaged drift

hypothesis for the actual finite coefficients. Notebook 33 identifies individually gauge-invariant diagonal determinant densities with the total Maxwell moment. Polynomial stationary jets on a constant Cartan-curvature probe remove the multiplier-curvature contribution through the required orders, including for the effective classical actions. A specified massive continuum proper-time reference has an explicit positive logarithm and bounded finite part in the same color and extraction convention. An exact heat integral for the actual lattice coefficient isolates sufficient estimates on three heat-time intervals. A closed-walk expansion bounds the actual based-ghost short-time contribution uniformly in endpoint and preceding depth. The vector contribution, the other intervals, and hence the sign for the actual family remain open. Notebook 34 computes the entire actual based-ghost coefficient. A returning-walk area formula, residue counting, the sharp point-pinned gap and a fourth-moment tail estimate give $C_H \log K/(12\pi^2) + O(1)$, uniformly in preceding depth. Its extra logarithm on heat times K^2 to K^4 proves that separated long-time ghost control at the usual block scale is impossible. The notebook also bounds the full actual vector short-time contribution, preserving the composite constraint. Thus the ultraviolet heat interval is closed; the intermediate and longer-time coupled comparison remains. An exact finite physical-quotient identity removes periodic pinning while retaining the fine/coarse gauge-orbit density ratio. The residual physical precision has a uniform flat gap cK^{-2}. A contour-gauge counterexample shows why the unprojected contour matrix cannot be assigned that gap. A paired scalar reference has exact infrared cancellation; its identification with the actual finite factors and uniform nonflat projected estimates remain open. Notebook 35 resolves the fixed-endpoint infrared passage by a new exact local gauge choice. A periodic flat-holonomy calculation first identifies a noncommuting scalar limit and its forced physical compensation. Moving-reference endpoint tangents then replace point pinning by zero covariant block means, with exact constant row Gram matrix. Finite Gaussian and Schur identities, continued jointly through flat stabilizers, express the actual determinant as a gauge-covariant scalar/vector pair. Both factors have flat gaps of order K^{-2}; a block-mean harmonic lift and a local tent lift prove full vector coercivity and Laplacian domination. Uniform heat and inverse-power bounds follow. Their local diagonal densities identify the actual Maxwell coefficient at every fixed endpoint. The large-endpoint curvature comparison and actual positive drift remain open. Notebook 36 factors that actual determinant into a soft fine scalar/vector pair and two coarse determinants. Both coarse Maxwell contributions are uniformly bounded as the endpoint grows. The proof controls physical source kernels without a fine-fiber dimension loss. A local gauge argument and an ordered Laurent factorization give uniform relative vertices for the actual effective-action Hessian; the vector matching uses its established on-shell affine jets. The remaining fine pair retains the effective action and its constraint terms. Both ends of its heat integral are now uniformly bounded: the current-time interval below one and the physical infrared tail above K^2. A local cubic-resolvent argument proves the latter without cyclically moving a block projector. The actual Maxwell logarithm is thereby reduced, up to a uniformly bounded term, to heat times $1 < s < K^2$. Notebook 37 closes this leading-determinant comparison under the explicit classical structural and local quantum inputs of the series. Source-localized sectorial resolvents remove the soft penalties on the whole band up to physical time one with an $O(u)$ heat error. The total nonlinear Ward jets through degree four then compare the actual effective-action base with the Wilson base with error $O(s^{-1})$, retaining the left-relative chart-current terms. A direct Wilson-to-continuum comparison has error $O(s^{-1/2})$, with its color and determinant normalization checked both by the magnetic heat kernel and by the transverse two-point diagrams. Consequently $\beta(\mathcal{D}_{M,K}) = 22\alpha_g \log K + O(1)$, uniformly in old depth and every applicable joint structural system. This supplies the positive margin required by finite moment transfer and local endpoint shooting. Arbitrarily long finite weak homogeneous local histories can be tuned to one fixed weak endpoint, with bare coupling of order the inverse square root of their length. These are research proofs from the stated upstream inputs, not external certification of the complete series or a continuum Yang–Mills existence theorem. Notebook 38 supplies exact positive good/remainder operators for a specified measurable completion of the local coarse map. Their iterates preserve the full Wilson mass and every history; local-germ agreement requires literal real-window containment and a possibly finite-volume-dependent orbit-overlap shrink. Separately, exact site-plane gluing gives uniform-volume bad-plaquette Peierls bounds under the normalized bare Wilson measure. A fixed-box conditional theorem includes its full compact normalization, and a boundary-only gauge joins these results into joint rarity of explicit local gauge/minimizer

window defects. These box windows are not yet the constrained windows of the multiscale map. Their identification, normalized large-component replacements, complete-density induction, and the continuum observable and reconstruction arguments below remain. Notebook 39 completes a restricted normalized replacement on arbitrary finite families of separated fixed boxes in the bare Wilson measure. It preserves the exterior marginal and raw mass, leaves bad-boundary laws unchanged, and gives exact sparse failure and coupling formulas. Its local total-variation, entropy, cluster and bounded-source cumulant comparisons are independent of the number of distant boxes. This finishes a finite local measure operation; it does not identify it with the constrained multiscale step or prove continuum existence.

Evidence boundary. The repository was examined at commit e392ef45013a581664bab76b90867170b30fe10e. The claims below concern the deposited material; unposted companion proofs have not been assessed. Elementary lemmas are proved here. Their applicability to the proposed Yang–Mills limit is always a separate assertion. No novelty or independent certification of the entire manuscript is claimed.

1 The target and the actual missing pieces

We should distinguish three milestones.

1. **Continuum scalar sector.** Construct a consistent family of separated Schwinger functions for a renormalized action-density observable, with reflection positivity, full Euclidean covariance, sufficient all-order distributional bounds, and nonzero positive-energy content.

2. **Yang–Mills theory.** Extend to an adequate family of gauge-invariant curvature polynomials and their derivatives, including mixed correlations and the correct tensor transformation laws. Identify their normalization, short-distance behavior, and identities along the Wilson continuum trajectory. Establish the appropriate vacuum-sector and reconstruction properties.

3. **The full existence and mass-gap problem.** In addition, prove a strictly positive gap and address every compact simple gauge group. This is stronger than the first two milestones.

The official problem statement explicitly asks for gauge-invariant local fields and appropriate short-distance behavior, together with the quantum field axioms and a mass gap [2]. One scalar called $\operatorname{tr} F^2$ does not automatically provide that observable family.

Obligation	Present status	Best next proof target
Uniform local correlations	Conditional source/RG inputs	One fixed physical stopping scale, a common vacuum normalization, and uniform complex-source bounds.
Subsequential existence	Follows conditionally from local bounds	Use a countable diagonal extraction; defer cutoff uniqueness.
All-order reconstruction bounds	Conditional factorial and joint-test reduction below	Verify the translated-grid source estimates and anchored polymer bounds that feed the reduction.
Full rotations	The supplied two-form theorem does not directly cover $\operatorname{tr} F^2$	Pass the two-frame positivity proved in Notebook 02 to the renormalized hierarchy, and prove an angular growth exponent strictly below four.
Nontrivial dynamics	The proposed lower witness is conditional	Compare two explicit reflected correlators at the same scale.
Yang–Mills identification and vacuum	Not supplied by a scalar witness	Construct mixed gauge-invariant observables and justify phase selection in the actual observable framework.

An ordinary positive probability law for the coincident composite Euclidean field is *not* a prerequisite we should impose without reason. The draft itself uses test functions flat on coincidence diagonals. An appropriate Osterwalder–Schrader reconstruction on that domain is a possible route, provided its quantitative growth assumptions are proved [3, 7]. Conversely, fixed-order separated correlations alone do not establish those assumptions.

2 A smaller source-stability interface

Write the finite-lattice generating function as

$$Z_{a,\Lambda}(w) = \int e^{-x_a A(U)+\mathcal{O}_a(w)}\, dU, \qquad \widehat{Z}_{a,\Lambda}(w) = e^{-C_{a,\Lambda}} Z_{a,\Lambda}(w).$$

Here a is the ultraviolet spacing and $C_{a,\Lambda}$ is a real, source-independent vacuum normalization. It cancels from all ratios. It is essential: the unnormalized bare free energy has a cost proportional to the number of *bare* sites. We require estimates proportional to the number of sites at one *fixed physical coarse spacing* h. Put $V = |\Lambda|/h^4$.

Lemma 2.1 (Elementary denominator floor)**.** *Suppose an exact coarse integral identity gives*

$$\widehat{Z}(0) = \int \rho(V_c)\, dV_c, \qquad \rho(V_c) \geq \chi(V_c) e^{-x_* A_h(V_c) - E_- V}, \quad 0 < x_* \leq X_*.$$

Suppose $\chi = 1$ when every coarse link lies in a fixed group ball B_δ about the identity, of normalized Haar measure $H_\delta > 0$. If $A_h \leq 6C_A\delta^2 V$ there, then

$$\widehat{Z}(0) \geq e^{-C_- V}, \qquad C_- = E_- + 6C_A X_* \delta^2 + 4\log H_\delta^{-1}.$$

All constants are independent of a and Λ.

Proof. Restrict the integral to that product of Haar balls. There are at most $4V$ links and $6V$ plaquettes. The restricted measure is at least H_δ^{4V}, and the asserted lower bound follows. □

The small-field characteristic function χ is a real hypothesis: in the RG construction it involves minimizing backgrounds, not merely the visible coarse plaquettes. Its uniform window must be verified. This elementary integration does not prove that window.

2.1 The fixed-cell reduction

Tile the coarse torus by M congruent reflection cells, each of volume v. Let $w_z = \sum_{i=1}^n z_i f_i$ be supported inside one cell, with a positive buffer from its faces. The f_i are real, with separated supports. Write $\nu_i = \|f_i\|_\sharp$ and $D_r = \{z : \sum_i |z_i|\nu_i < r\}$. Here $\|f\|_\sharp = \max(\|f\|_\infty, \operatorname{Lip} f)$ in the fixed coarse physical units. Let w_z^{ch} be its chessboard replication, conjugating coefficients on odd reflection copies.

Assume the following quantitative inputs.

1. The original Wilson measure gives

$$|Z(w_z)/Z(0)| \leq [Z(w_z^{\mathrm{ch}})/Z(0)]^{1/M}, \qquad Z(w_z^{\mathrm{ch}})/Z(0) \geq 0.$$

2. A local analytic source counterterm E_a, with $E_a(0) = 0$, obeys

$$\left| e^{-E_a(w_z^{\mathrm{ch}})} \widehat{Z}(w_z^{\mathrm{ch}}) \right| \leq e^{C_+ M v}.$$

3. Split E_a into cell-interior terms and face terms, each normalized to vanish at zero source. The latter obey $|E_a^{\partial}(w_z)| \le C_F$ and $M^{-1}|E_a^{\partial}(w_z^{\mathrm{ch}})| \le C_F$. There is a holomorphic recentering $E_a^{\#}(z)$, normalized by $E_a^{\#}(0) = 0$, with

$$\operatorname{Re} E_a^{\#}(z) = M^{-1} \operatorname{Re} E_a^{\mathrm{bulk}}(w_z^{\mathrm{ch}}) + \operatorname{Re} E_a^{\partial}(w_z), \qquad |\partial_1 \cdots \partial_n E_a^{\#}(0)| \le B_n \prod_i \nu_i.$$

One obtains the recentering by averaging the transported bulk terms, conjugating outside the terms from odd cells as well as inside their coefficients. This double conjugation is holomorphic.

Conditional theorem 2.2 (Uniform fixed-order correlations)**.** *Under these inputs and the denominator floor, set*

$$h_a(z) = e^{-E_a^{\#}(z)} Z(w_z)/Z(0), \qquad Q = e^{(C_+ + C_-)v + 2C_F}, \qquad r_0 = \frac{r}{2Q+1}.$$

Then $h_a(0) = 1$, $|h_a| \le Q$ on D_r, and $|h_a - 1| \le 1/2$ on D_{r_0}. In particular,

$$|\mathrm{cum}(\mathcal{O}_a(f_1), \ldots, \mathcal{O}_a(f_n))| \le \left[B_n + \log 2 \left(\frac{n(2Q+1)}{r} \right)^n \right] \prod_i \nu_i.$$

Proof. Combine the chessboard inequality with the source upper bound and denominator floor. The bulk real parts cancel exactly against $E_a^{\#}$; the two face errors cost at most $2C_F$. This proves $|h_a| \le Q$. On the complex line from zero through z, the Cauchy coefficient bound gives $|h_a(z) - 1| \le Qt/(1-t)$ for $t = r^{-1} \sum_i |z_i| \nu_i$. Hence a local logarithm has modulus at most $\log 2$ on D_{r_0}. Cauchy on radii $r_0/(n\nu_i)$, followed by the mixed derivative of $E_a^{\#}$, proves the result. □

The constants are independent of cutoff and full volume. A countable dense family of separated test tuples therefore admits a common convergent subsequence. Uniform multilinear bounds extend the limit to arbitrary separated tuples. Use cumulants of orders at least two to form moments of the *original* lattice observable centered by its real expectation. Reflection positivity passes through finite matrices of those moments. It is not asserted for cumulants or for the nonlinearly recentered generating function h_a. Cutoff uniqueness is not used.

One cell enclosing all insertions does *not* provide bounds uniform when two groups move arbitrarily far apart. For a contraction time semigroup one needs a per-ball version: finitely many occupied cells of fixed side, with the same face buffer, yielding a cost proportional to their number rather than the diameter of their union.

2.2 A checkable locality condition

At physical scales $\ell_j = L^{-j}$, one sufficient counterterm estimate is

$$\sum_{\substack{X^+ \cap B_R \neq \varnothing \\ \operatorname{diam} X^+ \ge t}} \sup_w |e_{a,j}(X; w)| \le A(R + \ell_j)^4 \ell_j^{-4} (1+j)^p e^{-ct/\ell_j}. \tag{2.1}$$

Each term must depend only on the source on X^+. The scale sum satisfies

$$H_p(t) := \sum_{j \ge 0} (1+j)^p L^{4j} e^{-ctL^j} \le C(1 + t^{-4})[1 + \log^+(1/t)]^p.$$

Split at $L^j \simeq 1/t$ and sum the geometric tail to prove this. A mixed derivative involving separated supports kills every term that does not meet them all. Surviving terms have diameter at least their separation. The same observation controls face errors using the buffer. Short local divergent terms are allowed; they must not be confused with the long terms whose sum is required to converge.

Status. This isolates a smaller verification task; it does not prove the source upper estimate or (2.1). Unsourced ultraviolet stability cannot be substituted for these source-dependent estimates. The deposited account already uses recentering and chessboard arguments; the reduction here clarifies the necessary interface and removes unnecessary uniqueness requirements.

2.3 The analytic estimate that the source restriction does not remove

The main source bound is supplied by the proposed correlation theorem in Raw II, lines 2791–2979, and its last-lattice argument at 59645–59855. It is not the later "source lemma" at 145173–145219, which is an additional input to the lower two-point witness. The bare density at lines 737–745 already includes a source-independent vacuum subtraction; that convention must be retained.

For a source f buffered from cell faces, let F_e denote its reflected copies and let F_o additionally change sign with reflection parity. Reflection and vanishing near the faces give $\|F_e\|_\sharp, \|F_o\|_\sharp \le \|f\|_\sharp$: split a line segment into its portions in individual cells and add their Lipschitz bounds. Thus

$$(zf)^{\mathrm{ch}} = (\operatorname{Re} z)F_e + i(\operatorname{Im} z)F_o$$

lies in the advertised two-profile source domain when $\sqrt{2}|z|\,\|f\|_\sharp < r$. This proves source-domain membership, not source stability. The independent complex decoupling variables used to localize the RG terms remain present even at zero source.

Dependency correction. The three localized links at Raw II, lines 48058–48320, describe site (L) in one part of the account. It has not been established that this reading is a required bottleneck of the active source proof. Its main induction consumers use a *global four-layer chain.* Raw I, lines 73107–73123 and 73545–73634, identifies that chain, explicitly excludes the alternative localized reading, and makes global carrier containment the base of the holomorphy argument. Its layers have both current and block-data inputs. The next model obligation is to verify this actual layer's joint domains, four-stage membership, and inherited or arrested slots. The following sufficient criterion is abstract: no identification of the global layers with these fixed-point equations is presumed.

Lemma 2.3 (Four-stage sufficient domain criterion)**.** *For field slots j, equip the full analytic chart norms with radii*

$$\delta_j = A_1 g_j (\log g_j^{-2})^{p_0}, \qquad \alpha_j = C_* g_j (\log g_j^{-2})^{q}, \quad q > p_0, \qquad \|u\|_\delta = \max_j \|u_j\|_j / \delta_j.$$

The norms must include all derivatives and other seminorms defining the charts. Put $\epsilon = \max_j \delta_j$. Suppose four response maps solve, successively,

$$H_i = G_i(\sigma_i)\xi_i + G_i(\sigma_i)\mathcal{N}_i(H_i; \sigma_i, \psi_i), \qquad i = 1, 2, 3, 4,$$

where each output becomes the next input. Assume jointly holomorphic coefficients on the required charts and decoupling polydisc, with

$$\|G_i\|_{\delta\to\delta} \le D, \quad D \ge 1, \qquad \mathcal{N}_i(0) = 0,$$

$$\|\mathcal{N}_i(u) - \mathcal{N}_i(v)\|_\delta \le A\epsilon(\|u\|_\delta + \|v\|_\delta)\,\|u - v\|_\delta .$$

Assume these bounds on the balls below, and that the background occupies at most half of every analytic radius. If

$$64AD^5\epsilon \le 1, \qquad 32D^4\delta_j \le \alpha_j \quad \text{for every } j,$$

then inputs $\|\xi_1\|_\delta \le 1$ give jointly holomorphic outputs of norm at most $2D, (2D)^2, (2D)^3, (2D)^4$, respectively. All outputs fit in the available analytic radii, uniformly in the number of slots.

Proof. For input norm s, use the ball of radius $R = 2Ds$. The fixed-point map has Lipschitz constant at most $2DA\epsilon R = 4AD^2\epsilon s$. Even at the largest input $s = (2D)^3$, this is at most $1/2$. Its linear part is bounded by $R/2$, and its nonlinear part by $DA\epsilon R^2 \le R/4$. It maps the ball into itself and contracts. Uniform Picard iteration gives holomorphy. Repeat four times and use $(2D)^4\delta_j \le \alpha_j/2$ for domain containment. □

At the same scale, a fixed small coupling ceiling γ supplies the required radius margin as soon as

$$(\log \gamma^{-2})^{q-p_0} \ge 32D^4A_1/C_*.$$

It also supplies small ϵ; no finite bound on the number of UV steps is needed. The difficult estimate is the *weighted* operator norm, for example the sufficient row bound

$$\sup_j \sum_\ell \|G_{j\ell}(\sigma)\| \frac{\delta_\ell}{\delta_j} \le D.$$

Ordinary spatial decay does not establish this when a current field is substituted into an arbitrarily old slot with a smaller analytic radius. The two-slot map $T(u_j, u_k) = (u_k, 0)$ has unweighted norm one, yet $T(0, \delta_k)$ escapes a radius $\alpha_j \to 0$. This is a diagnostic for the proposed estimate, not a counterexample to the actual localization maps.

There is a more specific positive result for one response chart actually displayed in Raw I. Its equations, at lines 14376–14437, have the form

$$A_0 = T[\Theta_J H_0 B - V'^J(A_0 + H_0 B)], \quad A = A_0 + H_0 B, \quad \mathcal{H}^J(B) = A - H^J D^J(A).$$

Direct differentiation gives

$$D_B\mathcal{H}^J = (I - H^J D'(A))(I + TV''^J(A))^{-1}(I + T\Theta_J)H_0.$$

The normalization includes $H_0 = G_0Q^*(QG_0Q^*)^{-1}\ell^{-1}$, where ℓ is the local layer-scale operator. It cannot be dropped. The coupling-flow inequalities in Raw I, lines 10234–10245, imply

$$\delta_k/\delta_j \le C_w\sqrt{1 + |k - j|}$$

once the fixed coupling ceiling is sufficiently small. For example, $x_j = g_j^{-2} \ge \gamma^{-2}$ and $-c_5 \le x_j - x_k \le B^\sharp(k - j)$ for $j \le k$ give this by bounding the square-root and logarithmic factors separately, with $B^\sharp\gamma^2 \le 1$ and $\log \gamma^{-2} \ge 2p_0$.

For this chart the multiscale metric obeys $d(y, y') \ge W(|j - j'| - 1)_+$, as recorded at Raw I, lines 13471–13483. Hence, if $\mu W \ge 1/2$,

$$e^{-\mu d(y,y')}\delta_{j'}/\delta_j \le \sqrt{2}C_w e^{-\mu d(y,y')/2}.$$

The weighted row bound therefore follows from the claimed response kernel estimate and block summability in that metric. In this geometry, scale-age weights are not an additional independent obstacle. The response kernel estimate itself and its full-polydisc antecedents still require verification. The B derivative alone does not cover the current input of the selected global layer. An accessible copy of the original CMP 122-II paper is now available; its formulas (1.51) and (1.54)–(1.55), on printed pages 370–371, are relevant to that global route. The alternative localized formulas (1.61)–(1.64) must not be substituted for it merely because some operator letters agree. Source extraction of several displayed formulas is incomplete, so equation-by-equation identification remains subject to that evidence limit. See [5].

The missing match is now more specific than an unnamed kernel bound. The global consumers at Raw I, lines 112589–112604, require five weighted output norms, including a graph norm containing $\ell^3|\Delta_{\mathcal{W}}\delta A'|$ (lines 111408–111435). The earlier response chart explicitly avoids applying covariant Laplacians to general independent-current responses (lines 15119–15141). A contraction in its weaker norm does not establish that stronger output estimate. On the input side, the field datum $\mathsf{j} = \widetilde{G}_0(\sigma)\mathsf{J}$ is itself assumed to satisfy a graph-norm bound at lines 72532–72535; treating it as arbitrary weak-norm forcing would also lose a hypothesis. The next proof must connect these actual input and output norms, or justify a revised interface that controls the combined current partner directly and still preserves every subsequent carrier domain. Identities valid only when the current is the configuration's actual Euler–Lagrange current cannot be used for an independent complex current.

There is also an upstream coupling-trajectory obligation. Raw I, lines 56419–56441, makes its clause-zero statement conditional on one-loop and unmarked-sector inputs. The concrete incomplete step at lines 4072–4088 transfers an unmarked remainder estimate to the chosen cutoff and radius; it is marked $(*)$. This is more precise than treating every cited one-loop coefficient as unproved. The effective-density representation is part of the joint induction. Thus the coupling-flow inequalities used in the weighted calculation above remain hypotheses for the construction. The original 1989 paper also states its stability theorem under an effective-coupling assumption; that historical statement does not by itself determine what later work has proved.

2.4 Two concrete repairs to test at the global interface

The explicit earlier chart can at least be extended to two inputs, without dropping the forcing. Freeze its background and decoupling coefficients, write G for its current-dependent Green operator, and put $L_0 = (I + G\Theta_J)H_0$. The equation

$$P = -\mathsf{j} + L_0 b - GV'(P)$$

extends the block-only chart of Raw I, lines 14387–14429. Assume V' is holomorphic on a neighborhood of the radius-$2r$ ball, $\|G\| \le M_G$, $\|V'(P)\| \le c\,\|P\|^2$ there, and $8M_G cr \le 1$. If $d = \|\mathsf{j}\| + \|L_0\|\,\|b\| \le r/2$, the right side maps the radius-r ball into the radius-$5r/8$ ball and has Lipschitz constant at most $1/2$. Indeed Cauchy gives $\|V''(P)\| \le 4cr$ there. Thus its unique small solution is holomorphic and $\|P\| \le 2d$.

For any weighted norm in which the same whole-ball Hessian bound is $\|GV''(P)\|_p \le 1/2$, set $R_P = (I + GV''(P))^{-1}$ and $T_P = I - H^J D^{J\prime}(P)$. The layer of this earlier chart is $\mathcal{H} = P - H^J D^J(P)$, and its exact data derivatives are

$$D_{\mathsf{j}}\mathcal{H} = -T_P R_P, \qquad D_b\mathcal{H} = T_P R_P L_0.$$

For $\mathsf{j} = Gf$, the forcing derivative is $D_f\mathcal{H} = -T_P R_P G$. These are genuine two-input formulas, conditional on the stated primitive and Hessian bounds; they do not identify this chart with every later global layer.

Still taking $\mathsf{j} = Gf$, at the undecoupled value $\sigma = 1$ use $G = (I - \bar{G}\Theta_J)^{-1}\bar{G}$ and the operator identities $\bar{G} = P_0 G_0$, $\Delta_a \bar{G} = P_0^T$ and $\Delta_a H_0 b = Q^* q_0 \ell^{-1} b$ give the forced representation

$$\Delta_a P = P_0^T[-f + \Theta_J P - V'(P)] + Q^* q_0 \ell^{-1} b.$$

This controls the augmented elliptic operator without differentiating f. It does not control the covariant Laplacian of $P - H^J D^J(P)$ automatically. Nor does walk decoupling preserve an inverse identity without an additional proof.

There is also an exact simplification of an actual global consumer. Let $L_\sigma = \Delta_W - P_1(\sigma) + \mathsf{K}$ and retain the deposited partner-current formula

$$\mathfrak{s}_\sigma^\sharp = L_\sigma A' - \mathfrak{m}_H(\sigma)X + F(W, \mathbb{H}).$$

For a scalar cutoff χ whose arguments stay in the domain of F, the cut current defined at Raw I, lines 111542–111543, satisfies

$$\begin{aligned}\mathcal{J}^\sharp[\chi] = \chi\mathfrak{s}_\sigma^\sharp + [L_\sigma, \chi]A' - [D_W^* D_W, \chi]H(\sigma)X \\ + F(W, \chi\mathbb{H}) - \chi F(W, \mathbb{H}).\end{aligned}$$

This follows simply by expanding the first two terms; the separate $\mathfrak{m}_H(\sigma)X$ terms cancel. Local second-order commutators require first derivatives of the field and up to two derivatives of χ. For a nonlocal kernel P_1, its commutator has kernel $P_1(x, y)(\chi(y) - \chi(x))$, so a weighted first-moment kernel bound is the relevant input.

Consequently this cut consumer can potentially be controlled by an *independent bound on the combined current*, plus commutator bounds, instead of separate Laplacian and $\mathfrak{m}_H X$ estimates. The existing estimate of $\mathfrak{s}^\sharp$ uses those separate estimates, so reversing it would be circular. The missing step is to identify and bound a substituted current expression directly, and then check all remaining consumers and carrier domains. The algebra above is a concrete repair route, not that completed identification.

2.5 Separate coefficient extraction from the cutoff estimate

There is a useful logical refinement. The fourth hypothesis of Raw I's unmarked-increment lemma, lines 4053–4057, already assumes the small remainder estimate with its factor $\varepsilon_1^{\mathrm{I}}$. The modified-cutoff theorem at lines 4170–4209 is the upstream claim that must supply it for the actual fluctuation integral. The following elementary extraction lemma shows what can be proved once that analytic estimate and the correctly normalized response bounds are available. It does not prove the change-of-cutoff estimate itself.

Lemma 2.4 (Uniform quadratic extraction from pointed remainders)**.** *Let $R_{X,z}(A)$ be holomorphic for $\|A\|_X < r$, with $z \in X$ for finite nonempty $X \subset \mathbb{Z}^4$, $d(X) \geq 0$, and*

$$\sup_{\|A\|_X < r} |R_{X,z}(A)| \leq A_0 \epsilon e^{-\kappa d(X)}.$$

Assume, in the same normalized field norm,

$$\|h_x\|_X \leq H e^{-m \operatorname{dist}(x,X)}, \quad \operatorname{diam} X \leq M(d(X) + 2), \quad \sup_z \sum_{X \ni z} e^{-\kappa_0 d(X)} \leq K_0, \qquad \kappa > \kappa_0.$$

Here $r, m, M > 0$, $\epsilon \geq 0$, and all other bound constants are finite. Put

$$a = \min\{m/2, (\kappa - \kappa_0)/(4M)\}, \qquad C_* = 4A_0 H^2 r^{-2} K_0 e^{4aM}, \quad S_j(a) = \sum_{u \in \mathbb{Z}^4} |u|^j e^{-a|u|}.$$

Then the absolutely convergent kernel

$$\Pi(x, y, z) = \sum_{X \ni z} D^2 R_{X,z}(0)[h_x, h_y]$$

satisfies

$$|\Pi(x, y, z)| \leq C_* \epsilon e^{-a(|x-z| + |y-z|)}.$$

In particular, any quadratic coefficient with the displayed pointed form obeys

$$\left|\sum_{u,z} u_\mu u_\nu \Pi(0,u,z)\right| \le 4C_* S_0(a) S_2(a)\epsilon. \tag{2.3}$$

All constants are independent of a scale index if the hypotheses are.

Proof. Apply two-variable Cauchy to $R_{X,z}(sh_x + th_y)$ at radii arbitrarily below $r/(2\,\|h_x\|_X)$ and $r/(2\,\|h_y\|_X)$, then take their limits. The mixed derivative is bounded by

$$4A_0\epsilon H^2 r^{-2} e^{-\kappa d(X)} e^{-m(\mathrm{dist}(x,X)+\mathrm{dist}(y,X))}.$$

Since $z \in X$, $|x-z| + |y-z| \le \mathrm{dist}(x,X) + \mathrm{dist}(y,X) + 2\,\mathrm{diam}\,X$. The chosen a leaves at least $\kappa_0 d(X)$ in the decay exponent; the diameter allowance contributes e^{4aM}. Sum over X. Finally write $u = z + (u-z)$ and use $|u|^2 \le 2|z|^2 + 2|u-z|^2$. Summing the product of the two exponentials gives $4S_0(a)S_2(a)$, proving (2.3). □

Thus coefficient summation is not the same missing assertion as stability of the analytic remainder under the chosen cutoff. To apply this lemma, the deposited remainder must have the pointed localization and response/domain normalization just assumed, uniformly in the scale. Those identifications must be checked explicitly. An ordinary global supremum estimate or a real-field tail probability does not supply them.

Lemma 2.5 (Correlated Gaussian cutoff cost)**.** *Let $X_b \in \mathbb{R}^D$ be centered jointly Gaussian blocks. Suppose the covariance of the joint vector $(X_b)_{b\in S}$ has operator norm at most σ^2, $\sigma > 0$, uniformly over the finite sets S under consideration. Then*

$$\mathbb{P}\{\|X_b\|_2 > R \text{ for every } b \in S\} \le \left[2^{D/2}\exp\{-R^2/(4\sigma^2)\}\right]^{|S|}. \tag{2.4}$$

No independence of the blocks is assumed. For $R(g) = c(\log g^{-2})^p$, $c > 0$, $p > 1/2$, and fixed $C_1 > 0$, $q \in \mathbb{R}$, the bracket in (2.4) is at most $C_1 g(\log g^{-2})^q$ for all sufficiently small g. That threshold can be chosen independently of a later small parameter ϵ.

Proof. On the event in (2.4), $\sum_b \|X_b\|_2^2 > |S|R^2$. The Gaussian exponential moment at parameter $1/(4\sigma^2)$ is

$$\det(I - \Sigma_S/(2\sigma^2))^{-1/2} \le 2^{D|S|/2}.$$

Markov's inequality proves (2.4), including singular covariances by continuity. With $s = \log g^{-2}$, the logarithm of the ratio of its bracket to $C_1 g s^q$ is

$$\tfrac{D}{2}\log 2 - \log C_1 - \frac{c^2}{4\sigma^2}s^{2p} + \tfrac{s}{2} - q\log s \longrightarrow -\infty.$$

This proves the uniform eventual comparison, without reference to ϵ. □

Consequently the numerical conditions $C_1 g(\log g^{-2})^{p_1} \le \varepsilon_1^{\mathrm{I}}$ and a growing logarithmic cutoff radius can pay the Gaussian cutoff factor in the same order of parameter choices. What is still needed is the *localized analytic* remainder estimate on the fixed complex background domain, including its normalization. A tail probability for a real Gaussian cannot replace that estimate.

The original CMP119 discussion also narrows the required scope: on printed p. 276, terms localized wholly inside the small-field region retain the earlier conditions; the altered space (3.43) is assigned to terms meeting its boundary [6]. The whole-lattice unmarked calculation has no such boundary-crossing terms. Its immediate obligation is therefore the changed real cutoff with the fixed complex-background

radii, rather than arbitrary layered boundary domains. Also, the deposited exponents obey $p_1 < p_0$ (Raw I, lines 1355–1363). The auxiliary box $C_1(\log g^{-2})^{p_1}$ therefore cannot automatically be identified with a domain containing the full real cutoff of order $(\log g^{-2})^{p_0}$. If the actual comparison needs $gR(g) \le \varepsilon_1^{\mathrm{I}}$, this can be imposed by a further coupling ceiling chosen after $\varepsilon_1^{\mathrm{I}}$; its sufficiency for the analytic argument must still be verified.

The preparatory-window relative boundary estimate in Raw II, lines 2376–2385 and 2809–2812, remains a separate antecedent. The contraction lemma does not prove it. Nor can real-source positivity replace the complex estimate: for a Gaussian of variance $\sigma_a^2 \to \infty$, subtracting the exact holomorphic counterterm $\sigma_a^2 z^2/2$ makes the residual identically one, but the bound $|Z(u+iv)| \le Z(u)$ still loses $e^{\sigma_a^2 v^2/2}$.

3 A factorial bound without a shrinking global zero-free disc

A many-cell bound Q^n inserted into the preceding zero-free radius can produce an artificial loss of order Q^{n^2}. The following exact combinatorial argument avoids it.

Lemma 3.1 (Coherent subset bound)**.** *For each finite index set I, let h_I be holomorphic on $|z_i|\nu_i < r$, with $h_I(0) = 1$ and*

$$h_I(z_B, 0_{I\setminus B}) = h_B(z_B), \qquad |h_B| \le Q^{|B|}.$$

The same Q, r are used for every subset. Then

$$|\partial_I \log h_I(0)| \le |I|!(2Q/r)^{|I|} \prod_{i\in I} \nu_i. \tag{3.1}$$

No zero-free polydisc of fixed radius is assumed.

Proof. Put $m_B = \partial_B h_B(0)$. Cauchy gives $|m_B| \le (Q/r)^{|B|} \prod_{i\in B} \nu_i$. The local logarithm at the origin satisfies

$$\partial_I \log h_I(0) = \sum_{\pi\in\mathcal{P}(I)} (-1)^{|\pi|-1}(|\pi|-1)! \prod_{B\in\pi} m_B.$$

If $S(n,k)$ is a Stirling number of the second kind, the sum of absolute coefficients is $\sum_k (k-1)!S(n,k)$. Counting ordered partitions by their positive block sizes gives $k!S(n,k) \le n!\binom{n-1}{k-1}$. The coefficient sum is therefore at most $n!2^n$. □

The restriction identity requires a *fixed* grid and a subtraction functional that does not change when a source coefficient is set to zero. For disjoint real supports,

$$\left\| \operatorname{Re} \sum_i z_i f_i \right\|_\sharp + \left\| \operatorname{Im} \sum_i z_i f_i \right\|_\sharp \le 4 \max_i |z_i| \nu_i.$$

Thus an estimate for arbitrary finite real Lipschitz source families, depending only on their combined source norm, can be applied pointwise to the two profiles $\operatorname{Re} w, \operatorname{Im} w$. This gives a product source domain with radius independent of n. The functional must be independent of its presentation as a sum. The raw account advertises that interface; its proof remains an obligation.

Centered moments also retain factorial growth: if $|\operatorname{cum}_k| \le k! A^k \prod \nu_i$ for every subset and $k \ge 2$, their exponential generating-function majorant is $\exp((At)^2/(1-At))$. Cauchy at $t = (2A)^{-1}$ gives, for example, $|S_n| \le n!(4A)^n \prod \nu_i$.

3.1 Keeping the cell side fixed

The existing placement construction allows cell size to grow with n, which loses the needed factorial control. The following repair keeps one grid for every subset of a selected patch tuple; different tuples may use different grids because the final cumulants are multilinear. The underlying source estimates are still hypotheses.

Conditional theorem 3.2 (All-order flat-test bound)**.** *Assume the fixed-scale source, denominator and locality estimates above, with the following uniformities. An admissible fixed cell side ℓ can be used for the many-cell chessboard inequality; the RG grid and reflection grid may be translated together on a mesh of size $O(a)$; all constants are uniform in these translates and source positions. The source counterterms are presentation-independent holomorphic functionals on one fixed complex norm ball, vanish at zero, and obey the anchored estimate (2.1) for unit balls, including reflected weights. The bulk subtraction is kept fixed when source coefficients vanish. Enlarged polymer supports that connect a buffered source to a face, or connect separated sources, have the corresponding diameter lower bound. Assume a common subsequential hierarchy of centered off-diagonal moments has been extracted.*

Then, on Schwartz tests flat to all orders on every pair diagonal, this hierarchy extends uniquely and satisfies

$$|S_n(\Psi)| \le C(n!)^{23} q_{15n}(\Psi), \qquad q_m(\Psi) = \max_{|\alpha|\le m} \sup_X (1+|X|^2)^{m/2} |\partial^\alpha \Psi(X)|. \tag{3.2}$$

Here C is independent of n. If the anchored estimate is uniform also for shrinking anchoring balls with its explicit $(R+\ell_j)^4$ factor, the factorial exponent improves from 23 to 19. Neither exponent is asserted optimal.

Proof. Patch geometry. Choose a smooth spatial partition of unity on mesh $\varepsilon_n = \min(1,\ell)/(64n)$ with patch supports of coordinate width $2\varepsilon_n$. For one selected patch from each source, at most $6n\varepsilon_n < \ell$ of the phases of an ℓ-grid are forbidden in each coordinate if a buffer $2\varepsilon_n$ is required. Choose allowed phases and round to the admissible grid mesh. For $a \le c\varepsilon_n$, every patch retains buffer $b_n \ge \varepsilon_n$. Keep this grid and its subtraction data for all subsets of this tuple. There are at most n occupied cells. For a source supported in a ball of radius R,

$$\sum_s \|\chi_s^{\varepsilon_n} f\|_{C^1} \le C(1+R)^4 n^5 \|f\|_{C^1}. \tag{3.3}$$

This follows from $O((1+R)^4\varepsilon_n^{-4})$ patches and the first-derivative cost $O(\varepsilon_n^{-1})$.

Face cost and source radius. For a holomorphic functional F on a norm ball of radius r_*, $F(0)=0$ and $|F| \le B$ imply $|F(w)| \le B\|w\|/r_*$ by radial Schwarz. Apply this to the absolutely summable face counterterms. It is legitimate even for a conjugate-reflected weight: freeze that weight and vary a complex scalar multiplying the entire weight, as allowed by the norm-ball hypothesis. Unit-ball anchoring bounds the face cost per patch by

$$J_n \le CH_p(b_n) \le Cn^4(1+\log n)^p.$$

Let C_0 absorb the fixed comparisons between the complete complex source norm, reflected weights, and $\max_i |z_i|\nu_i$. Take $r_n = r_*/[C_0(1+2J_n)]$. On this smaller product domain the remaining face costs are at most one per source. The fixed-cell cost is also constant per occupied cell. Thus $|h_B| \le Q^{|B|}$ with Q independent of n, and (3.1) applies with $r = r_n$.

For the mixed derivative of the subtracted local functional, locality kills every polymer not meeting all selected supports. Anchor at the first patch and use its diameter at least their minimum separation δ to obtain $CH_p(\delta) r_*^{-n} \prod_i \nu_i$. A transported bulk term contributes only if all variables belong to one

cell; in that case use the same estimate in each reflected copy and average. No volume factor remains. Combining this with (3.1), (3.3), and multilinearity gives, for any $\eta > 0$,

$$|\kappa_n(f_1, \dots, f_n)| \le [C_\eta(1+R)^4]^n (n!)^{10+\eta}[1 + H_p(\delta)] \prod_i \|f_i\|_{C^1} . \tag{3.4}$$

Here $n^{9n}n!$ supplies ten factorial powers and the logarithm fits in $(n!)^\eta$ with an exponential constant. The restriction $a \le c/n$ is needed only eventually for each fixed n, which suffices for a countable subsequence extraction.

Centered full moments. Only partitions with blocks of size at least two contribute. For $s \ge 1$,

$$\sum_\pi \prod_{B\in\pi} (|B|!)^s \le (n!)^{s-1} \sum_\pi \prod_{B\in\pi} |B|! \le 2^{n-1}(n!)^s.$$

The last bound counts ordered partitions by positive compositions. There are at most $n/2$ nonzero blocks and $1 + H_p(\delta) \le C\delta^{-6}$ for $0 < \delta \le 1$. Hence, for supports in arbitrary translates of unit balls,

$$|S_n(f_1 \otimes \cdots \otimes f_n)| \le C_\eta^n (n!)^{10+\eta} \min(1, \delta)^{-3n} \prod_i \|f_i\|_{C^1} . \tag{3.5}$$

Joint tests. This final step uses only a bound of the form (3.5) with factorial exponent s. Put $\epsilon_j = 2^{-j}$ and choose a fixed smooth nondecreasing η_0 equal to zero on $[0, 2]$ and one on $[4, \infty)$. Set

$$F_j(X) = \prod_{a<b} \eta_0(|x_a - x_b|/\epsilon_j), \quad \chi_0 = F_0, \quad \chi_j = F_j - F_{j-1}.$$

These sum to one off the diagonals. On the support of χ_j or its derivatives, all pairs are separated by $2\epsilon_j$ and, for $j \ge 1$, some pair is within $8\epsilon_j$. Localize each variable on mesh $\rho_j = \epsilon_j/64$. Enlarged one-variable chart supports are separated by at least ϵ_j. Fourier expansion on each rescaled product chart gives tensor factors of C^1 norm at most $C\rho_j^{-1}(1 + |m_i|)$. Applying $(1 - \Delta_{y_i})^3$ for each variable bounds the sum of the absolute Fourier coefficients times $\prod_i(1 + |m_i|)$ by C^n times derivatives of order at most six in each variable. The frequency sum converges because $\sum_{m\in\mathbb{Z}^4}(1 + |m|)(1 + |m|^2)^{-3} < \infty$.

Each pair cutoff receives at most twelve derivatives; each spatial bump receives at most six. The product rule therefore costs at most $C^n n^{12n}$, using only fixed bounded-order cutoff derivatives. For $j \ge 1$, move one point of a close pair onto the other. Taylor flatness of order $9n$ applied after at most $6n$ test derivatives gives

$$|\partial^\alpha \Psi(X)| \le C^n \epsilon_j^{9n} q_{15n}(\Psi) \prod_i (1 + |c_i|^2)^{-3}, \qquad |\alpha| \le 6n,$$

on the relevant chart. The segment moves only one coordinate a bounded distance, so the product weight is bounded by $C^n(1 + |Y|^2)^{3n}$ along it and is absorbed by q_{15n}. Chart positions sum with $\sum_{c\in\rho_j\mathbb{Z}^4}(1 + |c|^2)^{-3} \le C\rho_j^{-4}$. The net small-scale factor is therefore

$$\epsilon_j^{9n}\epsilon_j^{-3n}\epsilon_j^{-n}\epsilon_j^{-4n} = \epsilon_j^n.$$

Its dyadic sum is bounded; $j = 0$ needs no Taylor gain. Since $n^{12n} \le e^{12n}(n!)^{12}$, the result is $C_\eta^n(n!)^{22+\eta} q_{15n}(\Psi)$. Choose $\eta = 1/2$ and absorb the exponential into another half factorial power to obtain (3.2).

Spatial cutoffs followed by F_J approximate a flat Schwartz test in every Schwartz seminorm: arbitrary Taylor flatness beats all fixed cutoff derivative losses. Thus compact tests away from diagonals are dense, which proves existence and uniqueness of the stated extension. No contact-term extension to all Schwartz tests has been chosen.

Finally, if shrinking-ball anchoring is uniform, each patch has radius $R_n \asymp b_n \asymp n^{-1}$. Its face sum is bounded instead by

$$C \sum_{j \geq 0} (1+j)^p (1 + R_n/\ell_j)^4 e^{-c b_n/\ell_j} \leq C(1+\log n)^{p+1}.$$

Split at $\ell_j \asymp b_n$: the coarse sum has logarithmically many terms and the fine sum decays exponentially in L^j. This removes four factorial powers throughout, giving exponent 19. □

The form of (3.2) is the linear-growth regularity condition used in OS reconstruction; an explicit modern formulation appears in [4, Section 2.5, equation (22)]. That paper's two-dimensional construction supplies no Yang–Mills input here.

Reflection positivity also extends, provided the hierarchy came from one common centered sequence of reflection-positive lattice theories. First approximate compact joint tests in the strict ordered positive-time chamber by tensor sums supported in separated product boxes. Each finite OS matrix is an entrywise limit of positive lattice matrices. Then use the joint bound and density of interior compact tests in the Schwartz space flat at the chamber walls. Reflected products are flat on all collision diagonals, including the time-zero wall, so positivity persists by continuity. This does not supply a second reflection, full rotations, clustering, or nontriviality.

For the deposited one-frame torus geometry, a period $N = 2L^k$ and cell side $\ell = L^r$, $r \leq k$, give an even cell count $N/\ell = 2L^{k-r}$. Bare-step translations preserve the link-plane faces and their antipodal partners. Thus odd L is compatible with this one-frame placement; no power-of-two cell count is required. The remaining translation uniformity concerns the RG source estimates, not this elementary tiling. Notebook 02 handles the additional requirements of two-frame placement on the same odd-L regulator.

There is also a concrete obstruction to removing the face buffer casually. For a centered reflection-positive Gaussian pair (X_+, X_-) with variances A_a and covariance $B_a \geq 0$, Cauchy–Schwarz yields

$$|\mathbb{E} e^{zX_+}| \leq \exp\Big(\tfrac{1}{2} A_a \operatorname{Re} z^2 + \tfrac{1}{2} B_a |z|^2\Big).$$

Subtracting the exact holomorphic counterterm $A_a z^2/2$ leaves the bound $e^{B_a|z|^2/2}$, even though the actual normalized function is identically one. If B_a diverges, a new estimate is needed. The real part of a holomorphic function vanishing at zero cannot uniformly cancel a positive divergent $|z|^2$ term: its circle average is zero. Interface locality by itself therefore does not fix complex chessboard discontinuities.

4 Rotation restoration: a sharp conditional threshold

The supplied rotation theorem applies to closed two-form laws with Gaussian domination in an unweighted L^2 norm. It does not directly apply to the composite scalar $\operatorname{tr} F^2$. In particular the non-Abelian Bianchi identity is $d_A F = 0$, rather than $dF = 0$.

4.1 A finite-lattice ingredient can be proved directly

For the pure Wilson measure, diagonal reflection positivity itself has a short proof on the correct finite-volume geometry. Use the period lattice generated by $n(e_0 + e_1)$, $m(e_0 - e_1)$ and transverse coordinate periods. Reflection across $s = t + x = 0$ fixes the sheets $s = 0, n$ and separates the two open halves. Only (t, x) plaquettes cross a sheet. Their two length-two half-path holonomies are A and B, and their Boltzmann factor is

$$e^{c \operatorname{Re} \operatorname{tr}(AB^\dagger)}, \qquad c \geq 0.$$

In real and imaginary matrix coordinates, $\operatorname{Re}\operatorname{tr}(AB^\dagger)$ is an ordinary real inner product. Expanding its exponential in coordinate monomials gives a sum $\sum_\alpha c_\alpha a^\alpha b^\alpha$ with $c_\alpha \geq 0$. Condition on links lying in the fixed sheets and integrate the remaining links with product Haar measure. Each term is a squared half-space integral. This proves reflection positivity for bounded half-observables, including for $SU(2)$ and $SU(N)$. Diagonal Wilson positivity is also used explicitly in [10].

This does not settle *same-limit* positivity. Ordinary coordinate cubic tori have a different geometry. One must compare their selected limit with the adapted construction, or choose a construction retaining all required symmetries from the outset. Centered free-boundary cubes retain both central reflection directions, but would require a separate proof of translation invariance in their limit. Changing boundary conditions does not remove that obligation.

4.2 A compact-group mechanism for transferring insertions

The same-limit argument in companion Notebook 02 exploits the compact Wilson link variables. The following elementary lemma is its local analytic input; the full proof additionally needs the geometry, slab normalization, and limit argument given there.

Lemma 4.1 (Wilson convolution conjugation)**.** *On $G = SU(2)$ with normalized Haar measure and fixed $\beta > 0$, let*

$$k_\beta(g) = e^{(\beta/2)\operatorname{tr} g}, \qquad (C_\beta\psi)(v) = \int_G k_\beta(vu^{-1})\psi(u)\,du.$$

If f is a finite Peter–Weyl polynomial of largest spin r, then $T_f = C_\beta^{-1} M_f C_\beta$, initially defined on finite representation sums, extends continuously from $C^\infty(G)$ to itself. For each Sobolev index s,

$$\|T_f\psi\|_{H^s} \leq C_{\beta,f,s}\,\|\psi\|_{H^{s+2r}}\,.$$

The reverse conjugation obeys the same type of estimate. Tensor-product versions hold on any fixed finite set of links, with constants independent of the ambient volume.

Proof. The spin-j eigenvalue of C_β is $c_j = 2I_{2j+1}(\beta)/\beta > 0$. Indeed, integrate the character $\sin((2j+1)\theta)/\sin\theta$ against $(2/\pi)e^{\beta\cos\theta}\sin^2\theta\,d\theta$ and divide by $2j+1$. Multiplication by f connects only spins k with $|k-j| \leq r$. For integer $n \geq 0$,

$$I_n(\beta) = \frac{(\beta/2)^n}{n!}S_n, \qquad 1 \leq S_n = \sum_{l\geq 0}\frac{(\beta^2/4)^l n!}{l!(n+l)!} \leq e^{\beta^2/4}.$$

Thus ratios of the eigenvalues on this finite band grow at most as $C(1+j)^{2r}$. Each multiplication block has norm at most $\|f\|_\infty$, and only a bounded number of spin sectors are connected. Weighted square summation proves the Sobolev bound; Sobolev embedding gives continuity on C^∞. □

In particular $C_\beta T_f B = f C_\beta B$. Since the kernel is symmetric, this is the exact insertion identity

$$\int k_\beta(vu^{-1})f(u)B(v)\,dv = \int k_\beta(vu^{-1})(T_f B)(v)\,dv.$$

On a fixed set of crossing links it can transfer a lower-layer polynomial into a smooth function of the next layer. A local bounded smooth perturbation e^{tR} changes the resulting expression by $O(t)$ in supremum norm, by the finite derivative loss and compactness. These constants may depend on β; this is a fixed-lattice step.

Companion Notebook 02 completes the three additional interfaces at fixed $\beta > 0$: gauge-invariant layer removal using a conditional convolution identity on the original link law; a clean-plane operator estimate from exact physical transfer normalization; and two-frame placement using alternating

coordinate site/link walls. The latter retains the actual side $n = 2L^M$, L odd, which cannot satisfy the convenient QED3 divisibility assumption. Diagonal feature kernels give the exact cellwise folds. Full reflected source arrays are necessary in the first Jensen comparison to retain a pressure error divided by the volume. The assembled result is that every fixed-coupling coordinate-torus subsequential infinite-volume limit has diagonal RP. Its proof takes volume to infinity before removing the auxiliary tilt, uses convex separation rather than uniqueness, and has been internally checked but not externally certified. Passing this property to a renormalized continuum hierarchy still requires that hierarchy's common convergence.

4.3 A counterexample that survives the physical quotient

Let ϕ be a massive free Euclidean scalar in $\mathbb{R}^4$ and put $Y = P(\partial)\phi$, where $P(\partial) = \sum_{\mu=0}^{3} \partial_\mu^4$. Then

$$\widehat{C}_Y(p) = \frac{(\sum_\mu p_\mu^4)^2}{p^2 + m^2}.$$

This law has translations, signed coordinate permutations, factorial Schwartz bounds, and reflection positivity at coordinate and 45-degree mirrors. Indeed P is local and commutes with those reflections, so all their positive-half-space observables pull back to those of ϕ.

Nevertheless Y, declared to be a scalar, is anisotropic even off the diagonal. For a $\pi/4$ rotation R, a contact-only covariance difference would force $p^2 + m^2$ to divide $P(Rp)^2 - P(p)^2$. At $p = (z, 0, 0, 0)$ the latter equals $-3z^8/4$, nonzero at $z = im$. This disproves divisibility. The OS quotient is nontrivial: its one-particle multiplier is $\omega(k)^4 + \sum_j k_j^4 \geq m^4$. Thus broad regularity plus the two reflection directions is insufficient.

4.4 The angular-degree lemma

Let $M_N(T; \zeta)$ be an angular moment for an admissible N-tuple T. Let J denote antilinear time reflection. Assume:

1. M_N is entire, $\pi/2$-periodic, and

$$|M_N(T; \theta + i\beta)| \leq C_{N,T} e^{\kappa N|\beta|}, \quad 0 \leq \kappa < 4.$$

   Reality supplies both signs of every occurring frequency; $M_0 = 1$.

2. The tilted-frame moments obey $\widetilde{M}_N(T; \zeta) = M_N(T; \zeta + \pi/4)$.

3. For every real β, both kernels $M_{a+b}(J\chi, \eta; i\beta)$ and $\widetilde{M}_{a+b}(J\chi, \eta; i\beta)$ are positive semidefinite on positive-side tuples of *all* lengths, including zero.

4. If $r_N > 0$ is the largest absolute occurring frequency, the coefficient at frequency $-r_N$ is nonzero on some tuple $(J\chi, \eta)$ for every split $N = a + b$, including an empty side.

The last two requirements are analytic inputs, not consequences of reflection positivity at real angle alone.

Conditional theorem 4.2 (Strict angular type below four)**.** *Under these assumptions every M_N is independent of angle.*

Proof. Contour shifting kills Fourier frequencies $|\ell| > \kappa N$. Quarter-turn periodicity allows only $\ell \in 4\mathbb{Z}$. Let r_N be the largest absolute occurring frequency, zero if none. Detection and Cauchy–Schwarz at imaginary angle give

$$2r_{a+b} \leq r_{2a} + r_{2b}.$$

At order $2a$, multiply the positive kernels by $e^{-r_{2a}\beta}$ and let $\beta \to +\infty$. Their leading coefficient kernels differ by $e^{-ir_{2a}\pi/4} = (-1)^{r_{2a}/4}$. A nonzero kernel and its negative cannot both be positive. Detection therefore gives $r_{2a} \in 8\mathbb{Z}$. Set $b_j = r_{2j}$. It is convex, $b_0 = 0$, and its increments are nonnegative multiples of eight. Any positive increment forces eventual slope at least eight, contrary to $b_j \le 2\kappa j < 8j$. Thus $r_{2j} = 0$; the split $(0, N)$ gives $r_N = 0$. □

The threshold is sharp for this abstract argument. For the field Y above, each insertion has angular frequencies $0, \pm 4$, and the two-point function has a nonzero eighth harmonic: $(\cos^4\theta + \sin^4\theta)^2$ contains $\cos 8\theta$. Gaussian moments give maximal frequency $r_{2j} = 8j$ and type $4N$. The example also obeys the imaginary-angle kernel requirement: the differential operator $P_{i\beta}$ commutes with antilinear reflection, so free-field positivity applies to its transformed tests in either frame. The leading coefficients are analytic off coincidence and nonzero there; widely separated pairs retain their highest harmonics by massive decay. Permutation and translation then place a nonzero coefficient across each prescribed time split.

4.5 A free-field benchmark for the desired exponent

For the free four-dimensional Maxwell field, consider the separated Schwinger kernels of its Wick-ordered scalar action density $O = \sum_{\mu<\nu} :F_{\mu\nu}^2:$. Split coordinates into a plane (t, x) and two transverse coordinates y. Smear each insertion in y with $f_i \in C_c^\infty(\mathbb{R}^2)$. For an ordered complex plane difference z put $q(z) = \operatorname{Re} z_t - |\operatorname{Im} z_x| > 0$. The partial transverse Fourier transform of each field-strength covariance has the bound

$$|\widehat{K}_{\alpha\beta}(z, k_\perp)| \le C \int_{\mathbb{R}} \omega\, e^{-q(z)\omega}\, dk_x, \qquad \omega = (k_x^2 + |k_\perp|^2)^{1/2}.$$

The polynomial numerator from two derivatives is at most a constant times ω^2; the free spectral denominator is 2ω. Scaling and Plancherel imply

$$\|K_{\alpha\beta}(z, \cdot)\|_{L^2(\mathbb{R}^2)} \le C q(z)^{-3}. \tag{4.1}$$

For example, the momentum integral is bounded by $Cq^{-2}(1 + q|k_\perp|)e^{-q|k_\perp|/\sqrt{2}}$, whose $L^2(d^2k_\perp)$ norm is $O(q^{-3})$.

Every Wick graph for these composite insertions is a union of cycles of length at least two. The edge operator with kernel $f_i(y)K_{ij}(y - y')$ has Hilbert–Schmidt norm $\|f_i\|_2 \, \|K_{ij}\|_2$. The trace around each cycle is bounded by the product of these norms. The cone distance is superadditive: $q(z + w) \ge q(z) + q(w)$. Thus every nonadjacent ordered edge also has distance at least δ when successive gaps do. For $N \ge 2$,

$$|S_N(Z; f_1, \dots, f_N)| \le C_N \delta^{-3N} \prod_i \|f_i\|_2 \,. \tag{4.2}$$

This proves the exponent-three benchmark in a free model. It concerns separated kernels, not a claim that the coincident Euclidean Wick square is an ordinary random tempered distribution.

For imaginary planar rotations, $q(R_{-i\beta}z) \ge e^{-|\beta|}q(z)$. If the *interacting* ordered kernels satisfy the analogue of (4.2), and the necessary analytic gluing, detection and two-frame Gram identities are established, one obtains angular type $3N$. Controlled logarithmic losses can be absorbed into $(3 + \epsilon)N$, with $\epsilon < 1$. That would meet the theorem's threshold. Neither engineering dimension nor the free calculation proves this interacting bound. Notebook 02 supplies same-limit diagonal RP at fixed lattice coupling; its passage to the renormalized hierarchy and the interacting analytic/Gram inputs remain necessary.

5 A positive-energy witness at one coupling scale

Lemma 5.1 (Semigroup ratio)**.** *Let $H \geq 0$ be self-adjoint and $v \neq 0$. If $\langle v, e^{-tH} v\rangle \leq q \|v\|^2$ with $0 \leq q < 1$, then for $0 < \alpha < \log(1/q)$,*

$$\left\| \mathbf{1}_{(\alpha/t,\infty)}(H) v \right\|^2 \geq (1 - qe^{\alpha}) \|v\|^2 .$$

Moreover $(1 - e^{-tH})v \neq 0$, is orthogonal to every zero-energy vector, and has norm at least $(1 - q)\|v\|$.

Proof. On $[0, \alpha/t]$, $e^{-tE} \geq e^{-\alpha}$. Integrate against the spectral measure of v. The norm bound for $(1 - e^{-tH})v$ follows from Cauchy–Schwarz and its inner product with v. □

Here is a concrete test using the draft's own witness geometry. Set

$$f_d(x) = \left(1 - \frac{|x - (3d/4)e_0|^2}{(d/4)^2}\right)_+^2 .$$

It has supremum one, mass $\pi^2 d^4/3072$, and Lipschitz constant $32/(3\sqrt{3}\, d) < 7/d$. Thus its test-class parameter can be $\vartheta = 10^{-3}$. Both pairs $(\theta f_d, f_d)$ and $(\theta f_d, \tau_{4de_0} f_d)$ use the *same* scale d: their support separations are d and $5d$. Near-pair distances are at most $2d$, far-pair distances at least $5d$. For the exact continuum kernel $K(f, h) = \iint f(x)h(y)|x - y|^{-8}\, dx\, dy$,

$$\frac{K(\theta f_d, \tau_{4de_0} f_d)}{K(\theta f_d, f_d)} \leq (2/5)^8.$$

The draft's two witness prefactors have ratio twelve. Consequently, if its lower/upper witness and its sharp block-kernel asymptotic are proved,

$$\limsup_{d \downarrow 0} \frac{\langle v_d, e^{-4dH} v_d\rangle}{\|v_d\|^2} \leq 12(2/5)^8 = 0.00786432. \tag{5.1}$$

The running coupling cancels because the same d is used. One can then take $q = 0.02$ for sufficiently small d. With $\alpha = 1$, more than 94.5% of this vector's squared norm lies above energy $1/(4d)$.

5.1 Finite relative accuracy is enough

In fact the sharp asymptotic in (5.1) is stronger than needed. Let K_-, K_+ be the near and far continuum kernels, now with the common normalization $6/\pi^4$. If their discrete background counterparts obey

$$(1 - \epsilon)K_{\pm} \leq K_{\mathrm{bg},\pm} \leq (1 + \epsilon)K_{\pm},$$

the same witness gives

$$\frac{\langle v_d, e^{-4dH} v_d\rangle}{\|v_d\|^2} \leq 12 \frac{1 + \epsilon}{1 - \epsilon} (2/5)^8.$$

Already $\epsilon = 1/3$ gives $0.01572864 < 0.02$. Thus fixed relative accuracy on two shapes suffices for the stated positive-energy conclusion. The draft's coarse kernel constants alone do not supply this accuracy.

Here is an independent free-lattice calculation and the exact additional estimate that would supply it. In midpoint conventions put $s_\mu(q) = 2\sin(q_\mu/2)$ and $\lambda(q) = \sum_\mu s_\mu(q)^2$. The covariance multiplier of one colour of the exact two-form Gaussian is

$$m_{\mu\nu,\rho\sigma}(q) = \frac{\delta_{\nu\sigma} s_\mu s_\rho - \delta_{\nu\rho} s_\mu s_\sigma - \delta_{\mu\sigma} s_\nu s_\rho + \delta_{\mu\rho} s_\nu s_\sigma}{\lambda(q)}.$$

Write m_0 for the homogeneous multiplier obtained by replacing s by q, and $D_{PQ}(r)$ for its inverse Fourier transform away from zero. The potential covariance is normalized to $\delta_{\mu\nu}/(4\pi^2|r|^2)$.

Proposition 5.2 (Free off-diagonal approximation)**.** *For bare spacing a and plaquette-midpoint separation r with $|r| \geq C_0 a$, the infinite-lattice covariance satisfies*

$$|P_{\mathrm{ex},a}(p,q) - a^4 D_{\pi(p)\pi(q)}(r)| \leq C a^6 |r|^{-6}.$$

Proof. Near zero, $|\partial^\alpha (m - m_0)(q)| \leq C_\alpha |q|^{2-|\alpha|}$. A smooth dyadic decomposition into $|q| \asymp 2^{-j}$ therefore gives Fourier pieces bounded by $C_M 2^{-6j}(1 + 2^{-j}|z|)^{-M}$ for any fixed $M > 6$. Summing on both sides of $2^{-j}|z| = 1$ gives $C|z|^{-6}$. The lattice multiplier away from zero is smooth and periodic in the basepoint convention and has rapidly decreasing Fourier coefficients. For the continuum multiplier, sufficiently high derivatives of its high-frequency part are integrable; integration by parts gives the same rapid decrease away from contact terms.

For different orientations, apply the same midpoint phase $e^{iq\cdot(\delta_P - \delta_Q)}$ to both multipliers. Their difference still has order two at zero, and its integer Fourier coefficient at n compares kernels at $z = n + \delta_P - \delta_Q = r/a$. Finally use $D(r/a) = a^4 D(r)$. □

Suppose the actual background covariance has been identified as $C_{\mathrm{bg},a,h} = P_{\mathrm{ex},a} - C_{\mathrm{fl},a,h}$. The additional input needed is the uniformly normalized *curvature* bound

$$|C_{\mathrm{fl},a,h}(p,q)| \leq A a^4 h^{-4} e^{-c|r|/h}, \qquad |r| \geq h. \tag{5.2}$$

Here h is the physical witness-cell scale. A bound on the potential propagator alone does not give (5.2) without its discrete curl estimates. A scale decomposition with terms bounded by $A_0 a^4 h_j^{-4} e^{-c_0|r|/h_j}$, $h_j = hL^{-j}$, would suffice because

$$\sum_{j \geq 0} L^{4j} e^{-c_0 \rho L^j} \leq C_{L,c_0} e^{-c_0 \rho/2}, \qquad \rho \geq 1.$$

The elementary summation is proved by factoring out the right-hand exponential and using the summable majorant $\sum_j L^{4j} e^{-c_0 L^j/2}$.

For clarity, the discrete quadratic kernel meant here is

$$K_{\mathrm{bg},\pm} = \sum_{p,q} f_\pm(x_p) g_\pm(x_q) |C_{\mathrm{bg},a,h}(p,q)|^2,$$

with each positive plaquette orientation occurring once. The two covariance factors already supply the a^8 Riemann-sum weight. Any common colour or energy normalization must multiply both the continuum and discrete kernels.

Conditional theorem 5.3 (Two-scale criterion)**.** *Under the covariance identification and (5.2), the two fixed shapes satisfy the relative comparison with*

$$\epsilon \leq C_* \left\{ a/d + (d/h)^4 e^{-c\, d/h} \right\}, \tag{5.3}$$

for sufficiently small a/d, with a fixed reduction in c if needed. In particular finite thresholds $a/d \leq \eta_*$ and $d/h \geq R_*$ give $\epsilon \leq 1/3$.

Proof. All distances lie in $[d, 6d]$. The free approximation contributes $O((a/d)^2)$ against the curvature scale $a^4 d^{-4}$, and (5.2) contributes $O((d/h)^4 e^{-cd/h})$. Moving orientation midpoints to a common basepoint costs $O(a/d)$. For the six by six curvature matrix,

$$\|D(r)\|_{\mathrm{F}}^2 = \frac{6}{\pi^4 |r|^8}, \qquad |\, \|D+E\|_{\mathrm{F}}^2 - \|D\|_{\mathrm{F}}^2 \,| \leq 2\, \|D\|_{\mathrm{F}}\, \|E\|_{\mathrm{F}} + \|E\|_{\mathrm{F}}^2 .$$

Riemann sums for the fixed Lipschitz shapes cost $O(a/d)$ after dilation by d. Divide the absolute errors by the two fixed positive continuum kernels and absorb quadratic errors, reducing the exponential rate if necessary. This proves (5.3). □

The free finite-torus covariance tends to its infinite-volume value at fixed a by Fourier Riemann sums. This supplies no uniform torus error as $a \to 0$. Take volume first, or choose a sufficiently large volume diagonally for these two probes, and separately verify the conditioned covariance's volume limit. The deposited choice of h has d/h growing polylogarithmically as $d \downarrow 0$; it would provide the required separation once (5.2) is proved for that exact block map.

The interacting lower/upper witness and a valid reconstruction remain required. The result proves no mass gap: arbitrarily small positive energies in other states remain possible.

6 Why this still would not identify Yang–Mills

There is an explicit generalized-free countermodel to the logarithmic two-point signature. In four dimensions use positive spectral density

$$\rho(s) = \frac{as^2}{\log^2(e + s/\Lambda^2)}, \qquad C(r) = \int_0^\infty \rho(s) \frac{\sqrt{s}}{4\pi^2 r} K_1(r\sqrt{s})\, ds.$$

It defines a local generalized free Wightman field with positive energy, unique vacuum and clustering. Substitution $u = r\sqrt{s}$ and $\int_0^\infty u^6 K_1(u)\, du = 384$ give

$$C(r) \sim \frac{48a}{\pi^2} \frac{r^{-8}}{\log^2(1/(\Lambda r))}.$$

Thus this behavior does not by itself characterize non-Abelian dynamics. The general spectral construction is standard [8]; this choice and the displayed asymptotic provide the explicit counterexample.

Even a nonzero fourth connected function does not repair that criterion. Take a generalized free χ with $\rho_0(s) = a/\log(e + s/\Lambda^2)$ and its local Wightman Wick square $O =: \chi^2 :$. At separated Euclidean points,

$$C_\chi(r) \sim \frac{a}{2\pi^2} \frac{r^{-4}}{\log(1/(\Lambda r))}, \qquad C_O(r) = 2C_\chi(r)^2 \sim \frac{a^2}{2\pi^4} \frac{r^{-8}}{\log^2(1/(\Lambda r))}.$$

The connected four-point function of O is sixteen times the sum of the three distinct four-cycles of C_χ, and is nonzero. The forward-cone spectral measure has polynomial growth; Wick-square smearing is controlled by its positive-cone convolutions. A full operator-domain presentation would accompany any publication use. These are Wightman/separated-Euclidean examples, not unsubtracted Euclidean probability covariances at contact.

There is also a separate vacuum issue. Adding an independent constant random mode preserves the ultraviolet signature while creating degenerate zero-energy states. The temporal difference in Section 5 removes that contamination.

If an actual Euclidean probability measure with a local bounded observable algebra is available, phase selection has a useful route: translation-invariant events are measurable from either time half-space by translated Følner averages. Reflection positivity forces reflection to act trivially on that invariant algebra. Applying positivity to $\sqrt{h}A$, with h invariant and nonnegative, then gives positivity in almost every translation-ergodic component. After full Euclidean covariance, these components cluster; this follows from the relative Howe–Moore property of $(E(d), \mathbb{R}^d)$ for $d \geq 2$ [9]. The separated composite functions currently under discussion do not automatically provide that measure/algebra. An algebraic phase-selection argument or a direct clustering proof is still needed in their actual framework.

7 The next research sequence

1. **Close one sourced induction at a fixed stopping scale.** Use the joint fluctuation bounds of Notebook 17, the overlapping integration theorem of Notebook 18, and the physical Gaussian

and measure factors of Notebook 19. Notebook 20 closes their coefficient and precision match for the first bare Wilson small-field integral. Notebook 21 extends the local decomposition and one classical-reference integration uniformly through the minimized classical actions. Notebook 22 proves stationary-background substitution with explicit radius and support-weight losses. Notebook 23 proves local gauge invariance of the complete correction and its quadratic Ward identity. Notebook 24 gives the conditional inherited-interaction bound and an explicit sufficient next-step norm test. Notebook 26 satisfies that test with a persistent auxiliary-cell convention, returning one fixed positive support exponent for both inherited and new terms. Notebook 27 supplies an actual composite macrostep, including its exact local integral. Notebook 28 uses the smaller representative of Notebook 25 to return the same positive analytic radius at one step. It also contracts coefficientwise fifth-order background remainders and the quantum response to differences with matching low jets, at a sufficiently large fixed macro ratio and small coupling. Notebook 29 proves the actual local spatial symmetries, the total nonlinear Ward completion of the marginal term, and a connected localization of canceled low kernels with gain $C(J)L^{-\alpha}$ under its coherent-family and exact infinite-kernel hypotheses. Its classical semigroup permits the finite endpoint ratio $L = R^q$ to be chosen after fixing the elementary R, J. Its joint compactness and complete interaction quotient justify the classical change of representation under the stated map hypotheses. Notebook 30 establishes the universal classical gauge identity, canonical periodic descent and local continuity of the actual homogeneous quantum update. Its varying-input theorem identifies finite histories under their explicit intermediate-family budgets. Exact classical reference transport permits a projected finite-block contraction and an amplitude envelope on the selected infinite quotient while the couplings remain weak. Notebook 31 proves the corresponding finite local induction: finite Ward repair controls wrapping terms, and exact equivalent representatives meet every intermediate family budget. It also proves fixed-threshold coupling parity and exact preservation of the infinite Maxwell coefficient by classical substitution. On the infinite amplitude envelope the drift is the pivot/determinant coefficient plus a uniform error tending to zero with the coupling. The leading coefficient must still be computed. Notebook 32 gives qualitative uniform finite leakage, real-coupling continuity, an explicit convergent determinant trace, and an exact leading determinant cocycle. Its gauge/coarea identity supplies the actual coupled vector, ghost and coarse comparison terms, and bounds the shifted coarse effective-Hessian contribution uniformly in depth. A positive block margin in every varying-depth structural limit would transfer to finite regulators without a quantitative rate. Under a positive averaged drift hypothesis for the actual finite leading coefficients, its shooting argument produces arbitrarily long finite weak local trajectories with a fixed endpoint. Establish that drift through an actual uniformly controlled determinant comparison; it is not supplied by absolute increment bounds or the cocycle. Notebook 33 gives an exact constant-Cartan-curvature expression for the actual Maxwell coefficient and removes its multiplier curvature through second order using the classical envelope. Its diagonal trace densities have individual gauge identities, which are essential for matching the affine coefficient to the total Hessian moment. The notebook computes a specified massive continuum reference and rewrites the actual comparison as a heat integral. Uniform bounds on the short-time, intermediate and long-time parts remain to be proved, with the intermediate coefficient matched to the continuum value and the soft based-ghost modes controlled in the combined expression. The based-ghost short-time part alone now has a uniform bound from a closed-walk expansion on the actual pinned domain. Notebook 34 extends this to the entire based-ghost coefficient and proves its additional logarithmic interval from K^2 to K^4. It also proves the full vector ultraviolet bound using a uniform counting-ℓ^2 bound on the actual composite constraint and a localized trace on fixed coarse fibers. The short-time interval is therefore settled. The remaining vector estimate must match its intermediate coefficient and cancel the ghost's additional long-time logarithm, or supply

an equivalent bounded signed comparison. The same notebook gives an exact finite physical quotient, including both orbit-density determinants, and proves its uniform flat cK^{-2} gap. This supplies an alternative route to the combined comparison. Establish the joint infrared limit and the required curvature-source bounds before using the separate projected factors as infinite affine trace densities. Notebook 35 supplies that fixed-endpoint infrared bridge by changing the exact tangent gauge to covariant block means. It proves an actual local scalar/vector determinant identity, joint analytic continuation through flat stabilizers, individual gauge-invariant diagonal densities, and physical-scale flat gaps with uniform heat and inverse-power estimates for both new factors. Use this pair for the next large-endpoint curvature comparison. Its scalar projector, corrected tangent constraint, and full residual Hessian must be retained. The old point-pinned ghost logarithm cannot be assigned to the new scalar factor. The constant-flat anomaly found in the old physical quotient has canceled algebraically before the new local limits. Notebook 36 further removes the hard block projections by exact scalar and vector Schur factorizations. It proves that the two coarse determinant corrections have uniformly bounded actual Maxwell coefficients. Its physical relative-vertex theorem retains the true effective Hessian, using local path gauges and the ordered factorization of the flat kernel; the component-skew lattice terms are not discarded. Continue with the resulting fine soft pair and compare its marginal vertices and infrared penalties to a specified reference. The existing Ward extraction matches momentum jets through field degree four, but the required determinant remainder must still be bounded uniformly in preceding depth. The same notebook bounds the soft pair's ultraviolet and infrared ends, so it suffices to prove the actual intermediate heat coefficient $11\alpha_g$ with an error integrable in logarithmic time. Flat heat bounds alone do not identify this coefficient. Match the continued local representation to the measure, and control the complement of the small-field chart. Fix h, an endpoint inverse-coupling interval, the vacuum normalization, and the exact class of reflected source profiles. Prove the denominator window, the residual complex-source upper bound, and the anchored counterterm sum (2.1). Track every constant in a and $|\Lambda|$. The four-stage sufficient criterion in Section 2 is only an auxiliary tool. Audit the selected global two-input layer, its actual carrier domains and inheritance/arrest closure, together with the coupling trajectory and preparatory-window inputs. Reconcile the three-link localized description with that global route before treating its unresolved assumptions as prerequisites.

2. **Construct separated limits and preserve positivity.** Use the fixed-cell theorem and diagonal extraction. Prove the distance-uniform per-ball strengthening separately. Do not undertake two-cutoff uniqueness as a prerequisite.

3. **Make the all-order bounds usable.** Verify the translated-grid and presentation-independent holomorphy hypotheses of Section 3. Its patch argument, factorial lemma and joint-test extension then give the explicit linear-growth estimate (3.2). The remaining reconstruction assumptions, including symmetries and the vacuum condition, must be checked separately.

4. **Prove the interacting planar insertion estimate.** Use the free exponent-three calculation as a benchmark, not as a substitute. Use the same-state lattice positivity of Notebook 02, prove its passage to the renormalized hierarchy, and establish the analytic/Gram inputs of Section 4. The existing rotation-current route is an alternative only if its far-boundary flux condition is proved; renaming it does not establish it.

5. **Prove the two-shape kernel ratio.** Verify the constrained fluctuation-curvature bound (5.2), its volume limit, and the interacting witness. The free calculation and (5.3) then give one-third relative accuracy, already enough for positive-energy nontriviality through Section 5.

6. **Extend and identify the observable theory.** Introduce scalar, tensor and other gauge-invariant sources, control their operator mixing and joint limits, and match the required Yang–Mills short-distance structure. Prove a suitable vacuum-sector statement. Keep a mass-gap theorem as a distinct infrared task.

What this notebook has achieved. It reduces the first construction to explicit interfaces, proves several conditional steps and tests them against counterexamples. The largest unpaid debt remains the uniform *sourced* non-Abelian RG analysis. None of the deductions removes that debt.

Audit locations in the deposited account

These are navigation references to the pinned version, not assertions that every dependency at those locations has been verified.

- `theorem_0_complete_account_raw_III.tex`, lines 110862–110965: public proof-status ledger.
- Raw II, lines 60700–60765: local source terms, analyticity, stability and denominator floor; literal finite-family source domain.
- Raw II, lines 2791–2979 and 59645–59855: source correlation theorem and its final-scale consumer; lines 15206–15331 and 25412–25480 onward: sourced large-field step and joint induction.
- Raw I, lines 73107–73123 and 73545–73634: the selected global four-layer route and exclusion of the alternative localized chain; lines 56419–56441: conditional running-coupling input. The three-link discussion at Raw II, lines 48058–48320, is a distinct description of site (L); its relation to the active global-carrier route requires reconciliation before it identifies a necessary source gap.
- Raw I, lines 155039–155083: uniform characteristic-window proposal.
- Raw II, lines 63365–63367: outstanding final-scale statement; lines 63964–64274: per-ball recentering/chessboard reduction.
- Raw II, lines 145173–145222: outstanding source lemma and continuum-kernel proof.
- Raw III, lines 37365–37427: weak fine/coarse propagator stationarity.
- `ym4_manuscript_zeroth_draft.pdf`, Section 4.3, pp. 68–75: flat-diagonal test spaces and all-order growth issue; Definition 5.1 and Remark 5.7, pp. 77–78: witness geometry and claimed block-kernel asymptotic.
- `QED3_slop/YM_Rotation_Invariance_simplified.pdf`: Theorem A, p. 2; angular and two-frame argument, pp. 7–12; insertion bounds, pp. 13–16; contact lemma, pp. 17–18.

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